

Network Infrastructure

Item No.	Component	Purpose	Qty	Estimated Unit Cost	Estimated Total
1	T3 Internet Demarc Equipment	Provided by ISP; termination point for WAN	1	Provided by ISP	
2	Firewall Appliance (SMB-Grade)	Secures perimeter; stateful inspection; VPN	1	\$400	\$400
3	48-Port Gigabit Ethernet Switch	Core copper switching for all desktops + printers	1	\$450	\$450
4	4-Port Fiber Optic Switch	High-speed backbone to servers	1	\$600	\$600
5	Multimode Fiber Patch Cables	Fiber connections to servers	3	\$20	\$60
6	CAT7 Horizontal Cabling	Workstation & printer connections (avg 50 ft per drop)	35 runs	\$1/ft	\$1,750
7	Cable Installation & Termination	Professional CAT7 and fiber termination	1	\$5,000	\$5,000
8	Power Strips	Distributed in office and server room	35	\$15	\$525

Server Room Equipment

Item No.	Component	Description	Qty	Unit Cost	Total
9	SRV-APP01	Application/File Server (Fiber NIC)	1	\$1,200	\$1,200
10	SRV-DB01	Database Server	1	\$1,400	\$1,400
11	SRV-DC01	Domain Controller / DNS Server	1	\$1,000	\$1,000
12	UPS-1	UPS for SRV-DC01	1	\$150	\$150

13	UPS-2	UPS for SRV-APP01	1	\$150	\$150
14	UPS-3	UPS for SRV-DB01	1	\$150	\$150
15	UPS-4	UPS for Core Switch	1	\$150	\$150
16	UPS-5	UPS for Fiber Switch	1	\$150	\$150

Office Work Area Equipment

Item No.	Component	Description	Qty	Unit Cost	Total
17	Windows 11 Pro Desktop PCs	Primary employee workstations	30	\$550	\$16,500
18	PRN-EXEC-01	Executive Suite Printer	1	\$200	\$200
19	PRN-MKT-01	Marketing Printer	1	\$200	\$200
20	PRN-SALES-01	Sales Printer	1	\$200	\$200
21	PRN-ACCT-01	Accounting Printer	1	\$200	\$200
22	PRN-SHIP-01	Shipping/Receiving Printer	1	\$200	\$200

TOTAL ESTIMATED COST SUMMARY

Category	Total
Network Infrastructure	~\$8,760
Server Room Equipment	~\$4,250
Office Workstations & Printers	~\$17,300
Estimated Project Total	~\$30,310

Summary

To handle the building's backbone connectivity, this network architecture consists of a single small- to medium-sized business firewall appliance, a 48-port core Ethernet switch, and a 4-port fiber-optic switch. Dedicated UPS units safeguard three enterprise-grade servers (SRV-APP01, SRV-DB01, and SRV-DC01) that are connected via multimode fiber. Two other UPS devices support the network switches. The network is connected via CAT7 horizontal cabling to thirty Windows 11 Pro desktop computers and five

departmental printers (Executive, Marketing, Sales, Accounting, and Shipping/Receiving). The project cost includes a \$5,000 allowance for the expert installation and termination of all copper and fiber wiring. To ensure effective equipment placement and secure operations, power strips and structured wiring are installed throughout the server room and office.

Links

[Ubiquiti Store United States](#)

[TL-SG1048 | 48-Port Gigabit Rackmount Switch | TP-Link](#)

[MikroTik · CRS305-1G-4S+IN](#)

[Products overview | Schneider Electric](#)

[LC-LC OM3 Multimode Fiber Patch Cord Duplex 1m - FS.com](#)